

Autonomous Resilient Mesh Infrastructure for Distributed Edge Intelligence (ARMI-DEI)

NSF Program Solicitation: CISE / Robust Intelligence (RI) – Core Small

Lead Institution: Vertex Institute of Technology

Principal Investigator: Dr. Elena Rostova (Vertex)

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Proposed Duration: 36 Months (Oct 2026 – Sep 2029)

Total Requested Budget: \$598,420

Target Track: RI Small Collaborative Research

1 Project Overview and Research Motivation

Modern autonomous cyber-physical systems—ranging from autonomous aerial swarm networks operating in communication-denied environments to real-time smart grid power balancing systems—increasingly depend on decentralized decision-making at the network edge. Traditional centralized cloud architecture suffers from brittle network dependencies, excessive latency (> 100 ms), and catastrophic single-point failures when operating in dynamic, adversarial, or resource-constrained settings. While distributed machine learning (e.g., Federated Learning) has advanced edge analytics, current paradigms assume continuous topological connectivity and stable link bandwidth. They lack formal safety guarantees when network links undergo sudden degradation or Byzantine failures.

This project introduces **ARMI-DEI** (Autonomous Resilient Mesh Infrastructure for Distributed Edge Intelligence), a novel framework that unifies *neural-symbolic reasoning*, *dynamic topological consensus*, and *formal verification primitives* directly on heterogeneous edge hardware. By embedding lightweight formal logic engines into deep neural architectures, ARMI-DEI enables autonomous agents to perform provably correct, localized inference while dynamically reorganizing inter-node communication meshes under severe network partitioning.

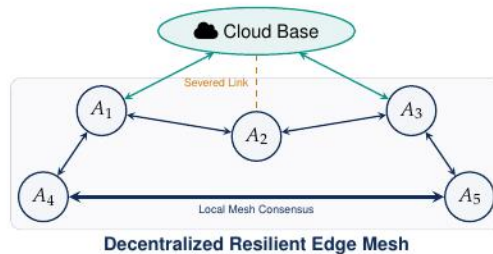


Figure 1: ARMI-DEI system architecture illustrating autonomous edge mesh reconfiguration during cloud disconnection.